

Decomposition

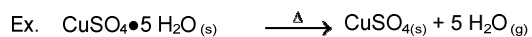
What is decomposition?

A chemical reaction in which a compound breaks down into two or more elements or simpler compounds.

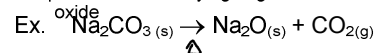
There are three main types of decomposition you should be familiar with:

I) Decomposition of a Hydrate:

- a hydrate is a compound with water in it

**II) Decomposition of a metal carbonate:**

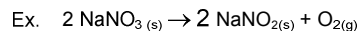
- products are always going to be CO_2 and the solid metal



<https://www.youtube.com/watch?v=8XoWj6H-iIM>

III) Decomposition of a metal nitrate:

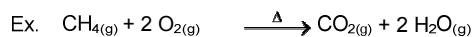
- products can decompose to a metal nitrite and oxygen gas



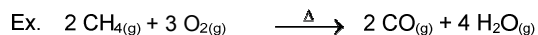
- Other decomposition reactions are also possible, depending on conditions

Combustion (2 Types)**I) Complete Combustion:**

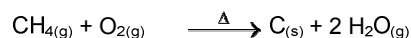
- there is an abundance of oxygen
- blue flame is produced
- products are carbon dioxide and water

**II) Incomplete Combustion:**

- there is not enough oxygen
- yellow flame is produced
- products are water, carbon monoxide and/or carbon (soot)



AND / OR



Read p122-136

Answer:

p125 #8, 10, 12

p127 #21-23

p132 #14, 16

p134 #31-33